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Exploiting hash functions to intensify the remote user authentication scheme

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ABSTRACT

Nowadays, the client–server model plays an important part in the Internet architecture. The procedure that a server authenticates a remote user securely has become a significant issue. Hence, many remote user authentication schemes are proposed. Hsiang and Shih pointed out the weaknesses of Yoon et al.'s scheme and proposed an improved scheme not only preserving original merits but also mending the weaknesses. The scheme protects against masquerading attacks, offline password guessing attacks and parallel session attacks. However, He et al. found that Hsiang et al.'s scheme is still vulnerable to password guessing attack, masquerading server attack and masquerading attack. Furthermore, we also found Hsiang–Shih's scheme could be threatened by a malicious insider that can originate an infringed account attack and a resembling account attack. In this paper, we first describe how a malicious insider carries out an infringed account attack and then present a resembling account attack on Hsiang–Shih's scheme. After that, we propose an improvement assisted by hashing functions to enhance Hsiang–Shih's scheme.

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1. Introduction

Nowadays, the client–server model plays an important part in internet architecture. Many remote user authentication schemes are proposed to allow a server to securely authenticate a remote user through an insecure channel. The most common scheme is the password authentication scheme that is to check the validity of the login message to authenticate remote users. The first password-based authentication scheme was proposed by Lamport in 1981. Thereafter, many modified schemes were proposed to improve the security, efficiency or cost (Yen and Liao, 1997; Lennon et al., 1981). In addition, there are many methods to implement password authentication scheme. In 2000, a password authentication scheme based on the collision-resist hash function was proposed by Peyravian and Zunic (2000). Later, Hwang and Yeh (2002) proposed an improvement by using the server's

public key. At the same time, Hwang and Li (Lee et al., 2002) proposed a new remote user authentication scheme using smart cards. Since then, many improvements were made to enhance its security and efficiency. Chien et al.'s scheme (Chien et al., 2002) features mutual authentication, no verification table and freely choosing password. Later, Ku et al.'s scheme (Ku and Chen, 2004) eliminates the vulnerability to reflection attacks and insider attacks. Yoon et al.'s scheme (Yoon et al., 2004) tried to protect against parallel session attack and improved the security. However, Hsiang et al. (Hsiang and Shih, 2009) pointed out the weaknesses of Yoon et al.'s scheme that is also vulnerable to parallel session attack, masquerading attack and password guessing attack. Thus, Hsiang et al. proposed an improved scheme to remedy these defects. Nevertheless, He et al. (2011) further found that Hsiang et al.'s scheme is still exposed to password guessing attack, masquerading user attack and masquerading server

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attack. Besides, we also find that an insider can carry out an infringed account attack and a resembling account attack on Hsiang et al.'s solution. In this paper, we first describe the attacks He et al. pointed out on Hsiang et al.'s scheme. Subsequently, we explain how an insider can launch an infringed account attack and a resembling account attack on Hsiang et al.'s solution. After that, we proposed an improved scheme that can fight against all the aforementioned attacks.

The rest of the paper is organized as follows: Section 2 briefly reviews Hsiang et al.'s scheme. Section 3 describes all the weakness of Hsiang et al.'s scheme. In section 4, we propose an improved scheme to against all attacks. We elaborate the cryptanalysis of the proposed scheme and compare the security of our protocol to other related protocols in section 5. The performance analysis is presented in section 6. At the end, section 7 concludes this paper.

2. Review of Hsiang et al.'s scheme

First we describe the notations used throughout this paper as in the follows:

- U : a user
- ID_u : user's identifier
- PW_u : user's password
- S : a remote server
- n : the number of times U re-register to S
- x : server's secret key
- T_u : user's current timestamp
- T_s : server's current timestamp
- $h(\cdot)$: a cryptographic hash function
- $\Rightarrow$: a secure channel
- $\rightarrow$: a common channel
- $\oplus$: bitwise XOR operation

There are four phases in Hsiang et al.'s scheme, the first is the registration phase, the second is the login phase, the third is the verification phase and the final is the password change phase. We describe these phases as in the follows.

2.1. Registration phase

In this phase, a user U initially registers or re-registers to a remote server S .

- 1) U selects his ID_u , PW_u and a random number b .
- 2) U computes $h(PW_u \oplus b)$ and sends ID_u , $h(PW_u)$ and $h(PW_u \oplus b)$ to S .
- 3) Upon receiving ID_u , $h(PW_u)$ and $h(PW_u \oplus b)$, S creates an entry in the account database and stores $n = 0$ in the entry for U 's first registration. Otherwise, S sets $n = n + 1$ in the existing entry for U . Then, S performs the following computations.
 $P = h(EID \oplus x)$, where $EID = (ID_u || n)$
 $R = P \oplus h(b \oplus PW_u)$
 $V = h(P \oplus h(PW_u))$
- 4) $S \Rightarrow U$: a smart card containing V , R and $h(\cdot)$.
- 5) U enters b into his smart card. Now the smart card stores V , R , b and $h(\cdot)$.

2.2. Login phase

In this phase, U performs the following operations to login to the server S .

- 1) U inserts his smart card into the smart card reader, and then inputs ID_u and PW_u .
- 2) The following computations are performed by U 's smart card.
 $C_1 = R \oplus h(b \oplus PW_u)$ ($= P$)
 $C_2 = h(C_1 \oplus T_u)$, where T_u denotes U 's current timestamp.
- 3) $U \rightarrow S$: $C = \{ID_u, T_u, C_2\}$.

2.3. Verification phase

After receiving the authentication request message $\{ID_u, T_u, C_2\}$, the remote system S executes the following operations.

- 1) If either ID_u or T_u is invalid or $T_s - T_u \leq 0$, then S rejects U 's login request. If not, S computes $C'_2 = h(h(EID \oplus x) \oplus T_u)$ and checks whether C'_2 equals C_2 or not. If $C'_2 = C_2$ holds, S accepts U 's login request and computes $C_3 = h(h(EID \oplus x) \oplus h(T_s))$, where T_s denotes S 's current timestamp. Otherwise, S rejects U 's login request.
- 2) $S \rightarrow U$: T_s, C_3 . After 1), S sends T_s and C_3 to U .

When receiving T_s and C_3 , the smart card first checks if either T_s is invalid or T_s equals T_u . If yes, U terminates this session. Otherwise, U computes $C'_3 = (C_1 \oplus h(T_s))$ and compares C'_3 to C_3 . If $C'_3 = C_3$ holds, U successfully authenticates S .

2.4. Password change phase

When U wants to change his password PW_u with a new password PW'_u , this phase will be invoked.

- 1) U inserts his smart card into the smart card reader, and requests to change password by entering ID_u and PW_u .
- 2) The smart card computes $P^* = R \oplus h(b \oplus PW_u)$ and $V^* = h(P^* \oplus h(PW_u))$.
- 3) The smart card verifies V^* with the stored V in the smart card. If the verification is passed, U successfully changes the password to PW'_u . Otherwise the smart card rejects the password change request.

The smart card then computes $R' = P^* \oplus h(b \oplus PW'_u)$ and $V' = h(P^* \oplus h(PW'_u))$. Next, it replaces R and V with R' and V' .

3. Attacks review

In this section, we first describe three attacks pointed out by He et al. and then we proposed the attacks on Hsiang et al.'s scheme. To evaluate the security of smart card based user authentication scheme, we assume that an attacker may have the following capabilities (He et al., 2011).

- (1) The attacker has total control over the communication channel between the user and the remote server. That is

the attacker may intercept, insert, delete or modify any message in the channel.

- (2) The attacker may either (i) obtain a user's password or (ii) extract the secret information of the smart card, but can't achieve both (i) and (ii) at the same time.

About the capability (2)(ii), by monitoring the power consumption (Messerges et al., 2002) or by analyzing the leaked information (Kocher et al., 1999), extracting the secrets stored in smart cards is relatively quick and easy. Although some smart card manufactures have taken the risk of these attacks into account and provided protections against the reverse engineering attempt. However, these smart cards are more costly. Thus, we would focus on the case that the attacker has capabilities (1) and (2)(ii).

3.1. Offline password guessing attack

After the registration phase of Hsiang et al.'s scheme, V , R , b and $h(\cdot)$ are stored in the smart card. We can see V , R , b and PW_u have the following relation among them.

$$\begin{aligned} V &= h(P \oplus h(PW_u)) = h(P \oplus h(b \oplus PW_u) \oplus h(b \oplus PW_u) \oplus h(PW_u)) \\ &= h(R \oplus h(b \oplus PW_u) \oplus h(PW_u)) \end{aligned}$$

According to this relation, the attacker A can perform the offline password guessing attack.

Step 1. A intercepts the user U 's ID_u over the communication channel in the login phase.

Step 2. A steals U 's smart card and obtains V , R and b from the by monitoring the power consumption (Messerges et al., 2002) or by analyzing the leaked information (Kocher et al., 1999).

Step 3. A performs the offline password guessing attack (Kai et al., 2009) as the following processes.

- 1) A chooses a password PW'_a from a uniformly scattered dictionary.
- 2) A computes $V' = h(R \oplus h(b \oplus PW'_a) \oplus h(PW'_a))$.
- 3) A verifies the correctness of PW'_a by checking whether V' equals to V or not.
- 4) A repeats step 1, 2 and 3 until V' equals to V .

Hence, PW_u can be obtained by the attacker A who launched an offline password guessing attack.

3.2. Masquerading user attack

After a successful offline password attack in 3.1, the attacker A can obtain PW_u and then A can use it to compute $P = R \oplus h(b \oplus PW_u)$. Thus, A can compute $C'_2 = h(P \oplus T'_u)$ to forge U 's login message and send ID_u , T'_u and C'_2 to S , where T'_u is A 's current timestamp. The forged message can pass S 's verification definitely.

Since ID_u and T'_u is valid, and $T_s - T'_u > 0$, where T_s denotes S 's current timestamp. S computes $C''_2 = h(h(EID \oplus x) \oplus T'_u)$ and checks whether C''_2 equals C'_2 or not. Because $C'_2 = C_2$ holds, S would accept A 's login request.

3.3. Masquerading server attack

Once the adversary A gets the value P by computing $R \oplus h(b \oplus PW_u)$, he/she can carry out the following steps to masquerade the server.

Step 1. A intercepts the message ID_u , T_u and C_2 in the verification phase.

Step 2. A computes $C'_3 = h(P \oplus h(T'_s))$, where T'_s is A 's current timestamp.

Then A can masquerade the server by sending the message $\{C'_3, T'_s\}$ to the user U . The message can successfully pass U 's verification. Since T'_s is valid and $T'_s > T_u$, U would compute $C''_3 = (C_1 \oplus h(T'_s))$ and compare C''_3 to C'_3 . Because $C''_3 = C'_3$ holds, U would consider A as S .

3.4. Infringed account attack

This attack can be performed easily by intercepting the ID_u in the login phase. Then the adversary A can utilize ID_u to execute the re-registration process.

Step 1. A chooses a random number b' and PW'_a , and then sends ID_u , $h(PW'_a)$, $h(b' \oplus PW'_a)$ to the remote server S .

Step 2. Since there is no any verification mechanism in the re-registration phase. S would set $n = n + 1$ and perform the computations in re-registration phase.

Step 3. A can get a new smart card issued by S , which sets $n = n + 1$, and it is easy to find that ID_u is infringed.

After this attack, the attacker A can acquire the whole information of the original user U . The original user U cannot use his/her smart card because n stored in his/her smart card is not equal to n stored in account database.

3.5. Resembling account attack

This attack can be performed under the condition of a special account format that is ID_u ended with numbers. As infringed account attack, the adversary A first intercepts ID_u in the login phase. Then A carries out the following step:

Assume the original user's $ID_u = \text{Bob921}$, then the user's EID contained in the smart card is $(\text{Bob921}||0) = (\text{Bob9210})$ and $P = h(EID \oplus x) = h(\text{Bob9210} \oplus x)$.

Step 1. Now since A knows the user's ID_u , A can make a registration toward the remote server S by a similar account $ID_a = \text{Bob92}$. Hence, A gets a new smart card containing $EID = (\text{Bob92}||0) = (\text{Bob920})$.

Step 2. Then A claims that he/she misses her/his smart card and re-registers to S . S will issue a new card to A and set $n = n + 1$ in the account database. Since there is no limitation in re-registration times, A will get a new smart card containing $EID = (\text{Bob92}||10) = (\text{Bob9210})$ after tenth re-registration. Also, A 's smart card contains $P = h(EID \oplus x) = h(\text{Bob9210} \oplus x)$.

Step 3. After getting P , A can carry out not only a masquerading user attack but also a masquerading server attack. That is A can compute $C'_2 = h(P \oplus T'_u)$ to forge U 's login message and send ID_u , T'_u and C'_2 to S , or compute

$C'_3 = h(P \oplus h(T'_s))$ and masquerade as the server by sending C'_3 and T'_s to U .

After this attack, the original user U can still use his smart card because the registration time of his account does not be modified.

4. Our proposed scheme

In this section, we propose our improved scheme that can protect against all the attacks mentioned in section 3.

Our scheme also has four phases, registration, login, verification and password change. The detailed procedures are described as follows.

4.1. Registration phase

In this phase, the initial registration is different from the re-registration. The process of the initial registration is depicted as follows.

- 1) User U chooses a random number b and computes $h(b \oplus PW_u)$.
- 2) $U \Rightarrow S: ID_u, h(PW_u), h(b \oplus PW_u)$.
- 3) In the account database, the server S creates an entry for U and stores $n = 0$ in this entry.
- 4) Next, S performs the following computations.
 $P = h(EID \oplus x)$, where $EID = (h(ID_u) || n)$
 $R = P \oplus h(b \oplus PW_u)$
 $V = h(h(PW_u) \oplus h(x))$ which is stored in U 's entry.
- 5) $S \Rightarrow U$: a smart card containing R and $h(\cdot)$.
- 6) When receiving the smart card issued by the server S , U inputs b into his smart card. Finally, the smart card contains b , R and $h(\cdot)$. After this phase, U does not need to remember b .

The whole procedure is depicted as Fig. 1.

If U misses her/his smart card and wants to re-register to S , the process of the re-registration is as the follows.

- 1) User U chooses a new random number b' and computes $h(b' \oplus PW_u)$.
- 2) $U \Rightarrow S: ID_u, h(PW_u), h(b' \oplus PW_u)$.
- 3) S computes $V' = h(h(PW_u) \oplus h(x))$ and compares V' with V .
- 4) If V' is equal to V , S sets $n = n + 1$ in the existing entry. Then S performs the following computations.
 $P_{new} = h(EID \oplus x)$, where $EID = (h(ID_u) || n)$
 $R_{new} = P \oplus h(b' \oplus PW_u)$
- 5) Finally, S performs steps 5) and 6) in the initial registration process.

Fig. 2 describes the re-registration phase.

4.2. Login phase

When U wants to login to the remote server S , the following operations will be performed.

- 1) U inserts his smart card into the smart card reader and enters his ID_u and PW_u .
- 2) The following computations are performed by U 's smart card.
 $C_1 = R \oplus h(b \oplus PW_u)$
 $C_2 = h(C_1 \oplus T_u)$, where T_u denotes U 's current timestamp.
- 3) $U \rightarrow S: C = \{ID_u, T_u, C_2\}$.

We present the login phase in Fig. 3.

4.3. Verification phase

After receiving the login request message $\{ID_u, T_u, C_2\}$, the remote server S and U 's smart card perform the following operations.

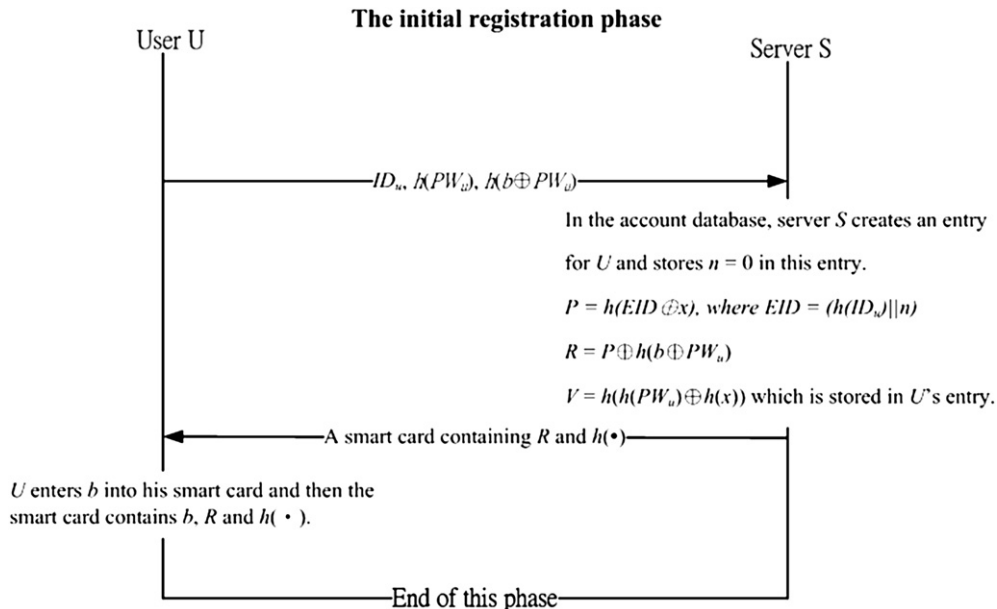


Fig. 1 – Registration phase.

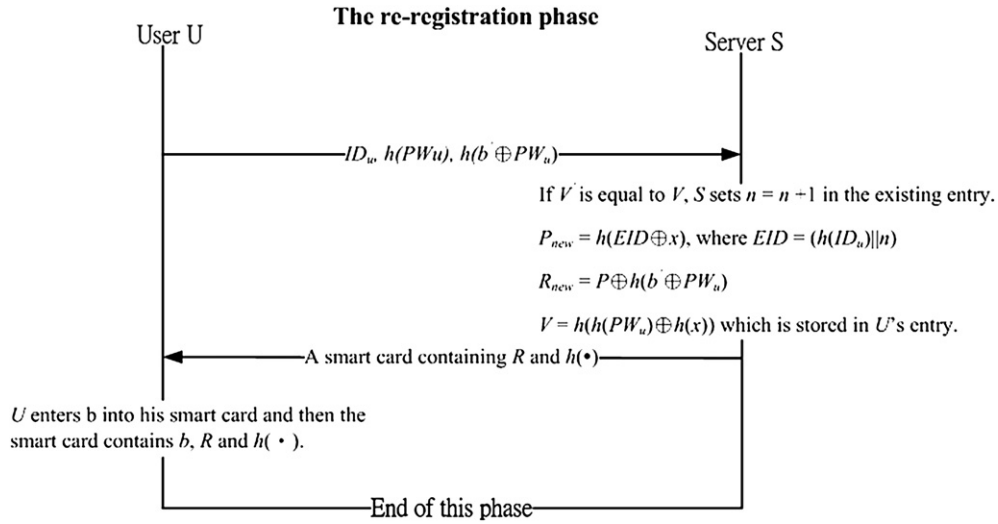


Fig. 2 – Re-registration phase.

- 1) If either ID_u or T_u is invalid or $T_s - T_u \leq 0$, S rejects U's login request. Otherwise, S computes $C'_2 = h(h(EID \oplus x) \oplus T_u)$ and compares C'_2 with the received C_2 . If $C'_2 = C_2$, S accepts U's login request and computes $C_3 = h(h(EID \oplus x) \oplus h(T_s))$, where T_s is S's current timestamp. Otherwise, S rejects U's login request.
- 2) $S \rightarrow U: T_s, C_3$.
- 3) If either T_s is invalid or $T_s = T_u$, this session will be terminated by U. Otherwise, U computes $C'_3 = h(C_1 \oplus h(T_s))$ and compares C'_3 with the received C_3 . If C'_3 is equal to C_3 , U authenticates S successfully.

The verification phase is shown as Fig. 4.

4.4. Password change phase

When U wants to change his password, the following process will be performed.

- 1) U inserts his smart card into the smart card reader and enters ID_u , PW_u and new password PW_{new} .
- 2) $U \Rightarrow S: ID_u, h(PW_u), h(PW_u) \oplus h(PW_{new}), h(b \oplus PW_{new})$.
- 3) S computes $V' = h(h(PW_u) \oplus h(x))$ and compares V' with V in the account database.

- 4) If V' is equal to V, then S computes $h(PW_u) \oplus h(PW_{new}) \oplus h(PW_u)$ to get $h(PW_{new})$. Next, S performs the following computations.

$$P = h(EID \oplus x), \text{ where } EID = (h(ID_u)||n)$$

$$R_{new} = P \oplus h(b \oplus PW_{new})$$

$$V_{new} = h(h(PW_{new}) \oplus h(x)) \text{ which is stored in U's entry.}$$

- 5) $S \Rightarrow U: R$.

- 6) Finally U's smart card replaces R with R_{new} .

Finally, Fig. 5 presents the password change phase.

5. Cryptanalysis of the proposed scheme

In this section, we describe the enhanced security features of our proposed scheme which is a modified form of Hsiang et al.'s scheme. In Yoon et al. (2004) and Hsiang and Shih (2009), the security analysis has been discussed and demonstrated. Therefore, this section will only discuss the improved security features of the proposed scheme. Our improved scheme not only preserves the merits of Hsiang et al.'s scheme such as providing mutual authentication, freely choosing password and parallel attack resistance but also protects against the attacks mentioned in section 3. The

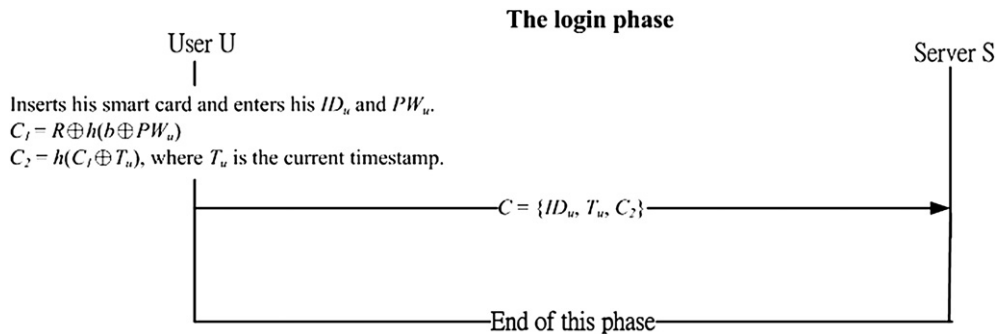


Fig. 3 – Login phase.

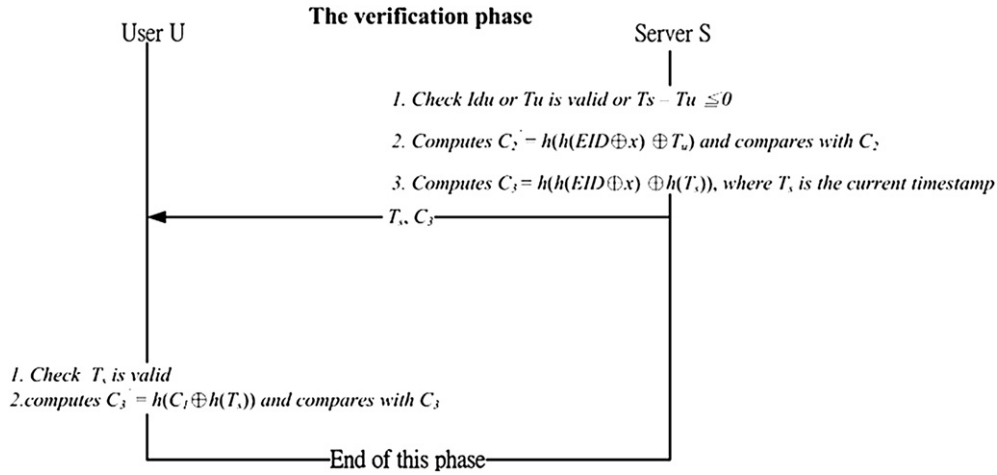


Fig. 4 – Verification phase.

security comparison of our protocol and other related protocols is presented in Table 1.

5.1. Offline password guessing attack

Our improved scheme removes the following relation between V , R and b about PW_u .

$$V = h(R \oplus h(b \oplus PW_u) \oplus h(PW_u))$$

Therefore, an attacker has no way to carry out the offline password guessing attack mentioned in 3.1.

5.2. Masquerading user attack

Since the offline password guessing attack proposed by He et al. cannot work on the improved scheme, an attacker cannot obtain $P = h(EID \oplus x)$ to forge U 's login request. Therefore, the improved scheme can successfully fight against the masquerading user attack.

5.3. Masquerading server attack

It is clear that an adversary cannot obtain P in password guessing attacks, thus he/she has no way to compute $C_3' = h(P \oplus h(T_s))$ to forge the remote server S . Hence, the improved scheme can prevent masquerading server attack.

5.4. Infringed account attack

If an adversary A successfully intercepts ID_u in the login phase, he/she cannot carry out an infringed account attack. Since in the re-registration phase of the proposed scheme, A must enter PW_u if he/she wants to utilize ID_u to re-register to the remote server. However, A does not know PW_u and therefore this attack is avoided.

5.5. Resembling account attack

If an attacker A wants to apply for a resembling account to implement this attack, he/she will try in vain. The improved

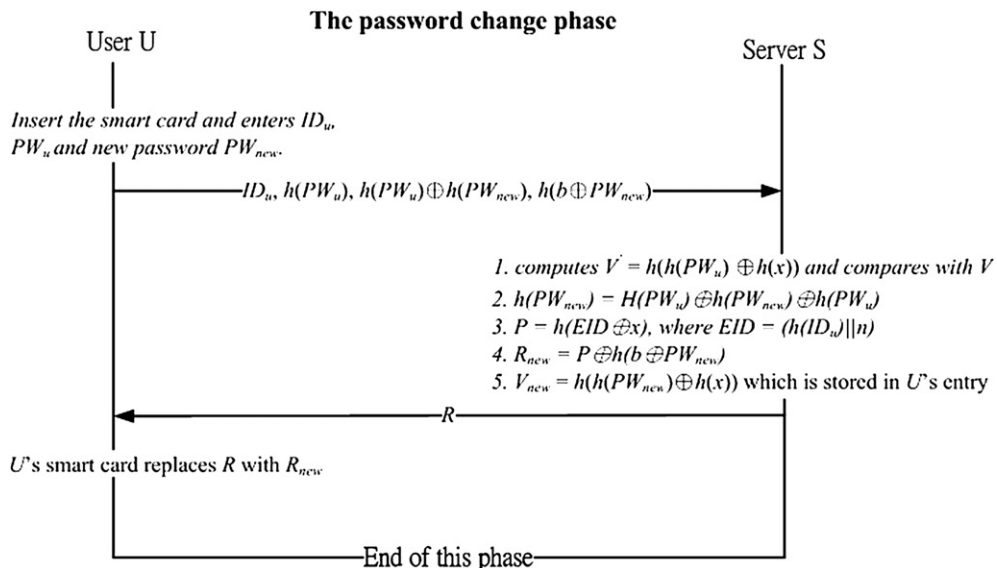


Fig. 5 – Password change phase.

Table 1 – Security comparison of our scheme and other related schemes.

Security features	Proposed scheme	Hsiang and Shih (2009)	Yoon et al. (2004)
Mutual authentication	Yes	No	No
Resist parallel session attack	Yes	Yes	No
Resist online password guessing attack	Yes	Yes	Yes
Resist offline password guessing attack	Yes	No	No
Resist masquerading server attack	Yes	No	No
Resist masquerading user attack	Yes	No	No
Resist infringed account attack	Yes	No	No
Resist resembling account attack	Yes	No	No

scheme uses a hash function to remove the intuitive value of $EID = (ID_u || n)$. For example, the accounts, Bob921 and Bob92, have different results in new-defined $EID = (h(ID_u) || n)$, since the hash values of Bob921 and Bob92 are distinct.

5.6. Online password guessing attack

Since in the re-registration phase, the remote server S will compute $V' = h(h(PW_u) \oplus h(x))$ to verify the correctness of PW_u . An attacker who intercepts an account ID_u may try different passwords, PW'_a , to pass the verification. However, this can be protected easily by restricting re-registration time.

5.7. Stolen verifier attack

If an adversary A steals user's information in the account database, he/she cannot carry the stolen-verifier attack because the password PW_u in $V = h(h(PW_u) \oplus h(x))$ is well-protected by hash functions and the permanent secret x of the remote server, which A does not know.

6. Performance analysis

An efficient authentication scheme must take computation cost into consideration during user's authentication. The comparison of the proposed scheme with other related smart card based authentication schemes is summarized in Table 2. The computation cost of the XOR operation can be ignored when compared to other operations. T_{hash} is defined as the time complexity for hash function. We mainly consider the computation of the login phase and the verification phase. Because these two phases are the principal parts of an authentication scheme and should be implemented for each session. In Table 2, it is clear that our proposed scheme needs the same computation cost as Hsiang et al.'s scheme and only

Table 2 – Performance comparison of our scheme and other related schemes.

Scheme	Login phase	Verification phase	Total
Proposed scheme	$2T_{hash}$	$6T_{hash}$	$8T_{hash}$
Hsiang et al.'s scheme (Hsiang and Shih, 2009)	$2T_{hash}$	$6T_{hash}$	$8T_{hash}$
Yoon et al.'s scheme (Yoon et al., 2004)	$2T_{hash}$	$4T_{hash}$	$6T_{hash}$

requires two more hashing operations than Yoon et al.'s scheme. However, the proposed scheme can achieve more security features which were not provided in their schemes.

7. Conclusion

In this paper, we present a cryptanalysis of Hsiang et al.'s scheme by showing that their scheme is vulnerable to password guessing attack, masquerading user attack, masquerading server attack, infringed account attack and resembling account attack. Subsequently, we propose an improved scheme and discuss the enhanced security features that provide a well protection against aforementioned attacks and suspicious attacks. We also compare the proposed improved scheme with existing schemes of the same type regarding both efficiency and security. The comparison results show that our improved scheme is more efficient and secure. Therefore, after continuous improvements of the authentication scheme, our scheme is more reliable and robust for real-life use.

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